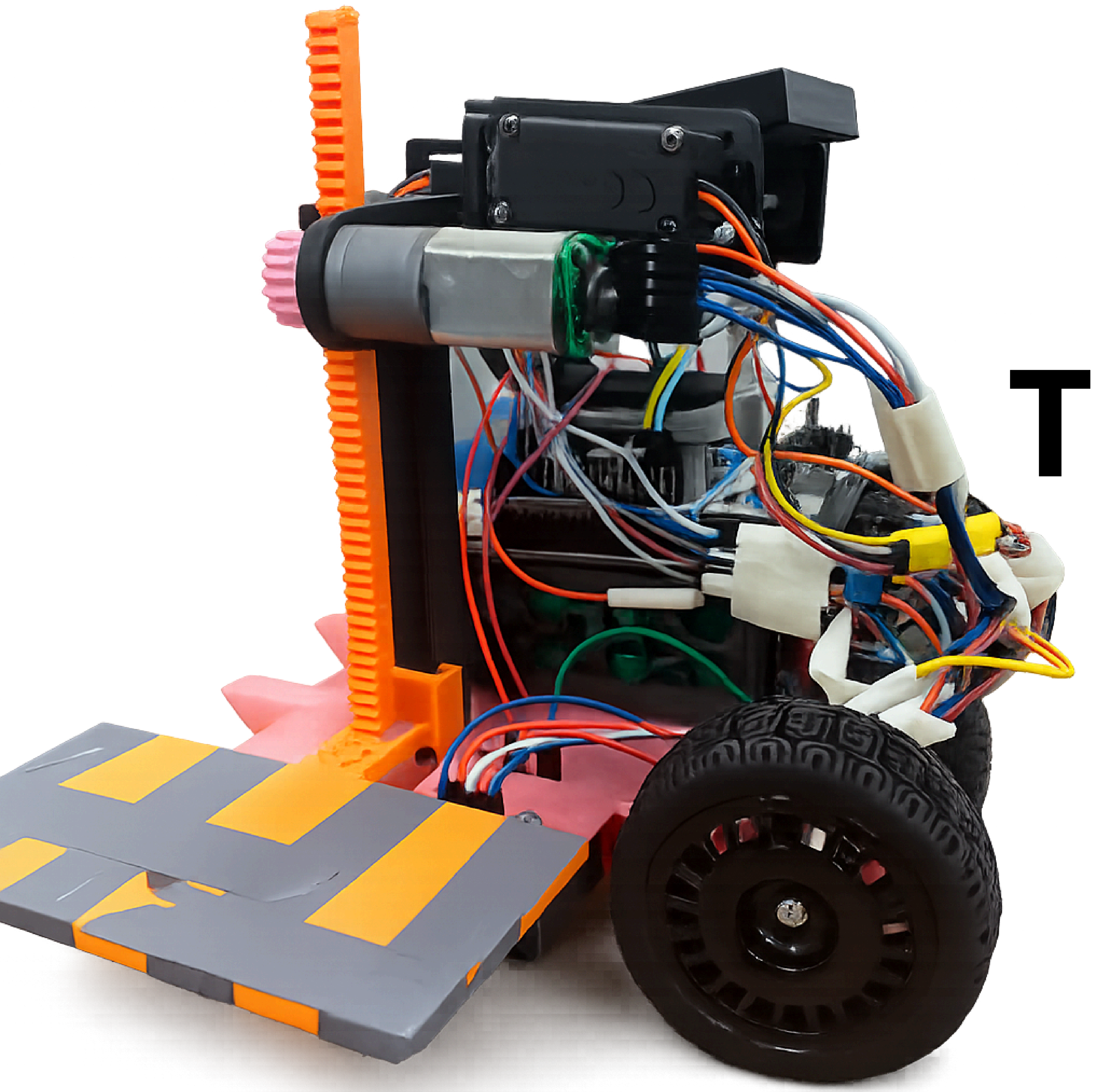
ASHESI UNIVERSITY

Zurich University of Applied Sciences

zh aw School of Engineering IMS Institute of Mechatronic Systems

MECH 617 Applied Systems Engineering



THE TRANSPORTER

Team:

- Alberta Petiafo
- Priscilla Yeboaa Asiedu
- Samuel Kojo Akwensivie
- Godlove Bissaga
- Douglas Kumi Koomson

PROJECT BRIEF

Mission Objectives

Build a robot that:

- Start at the depot.
- Follow the line clockwise.
- Identify package locations using color markers.
- Pick up packages.
- Cross the steps
- Deliver each package to the matching destination.
- Balance the ping-pong ball throughout the run.
- Finish in the shortest possible time.

Constraints

- Use only the provided hardware (STM32F446RE Nucleo and PES board, sensors, motors, etc.)
- Robot must have maximum dimensions of 200 x 200 x 200 mm at the start.
- Robot must use the supplied 12V NiMH battery pack
- Robot must operate autonomously and travel clockwise
- Robot must go over the step and cannot turn around
- *LineFollower.cpp* and the 1D Mahony filter must be developed by the team.

Challenge 1 - Crossing the step

- Design a wheel and chassis system that can climb a step
- **Steps at:**
  - 10 mm - Trainee
  - 20 mm - Engineer
  - 30 mm - King

Challenge 2 - Balancing the Ping-pong ball

- Use a 2DOF gimble mechanism with IMU to balance the ping-pong ball

Challenge 3 - Line follower

- Use the IR line sensor to follow the 20mm black line.

Challenge 4 - Detecting colour markers

- Use the colour sensor to identify the colour types [RED, BLUE, GREEN, YELLOW]

Challenge 5 - Picking packages and delivering at colour positions

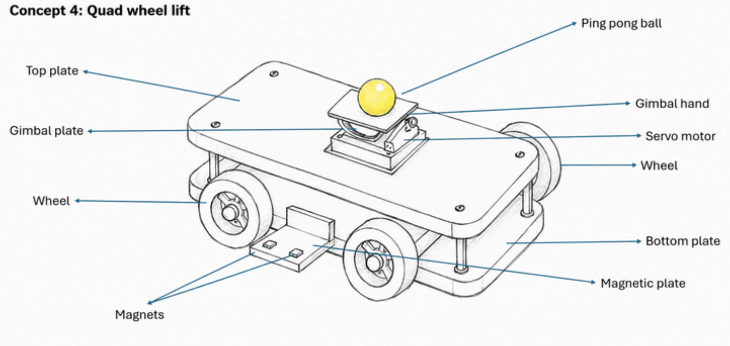
- The packages are [RED, BLUE, GREEN, YELLOW], pick them at the right position and deliver them correctly

Main Inventory

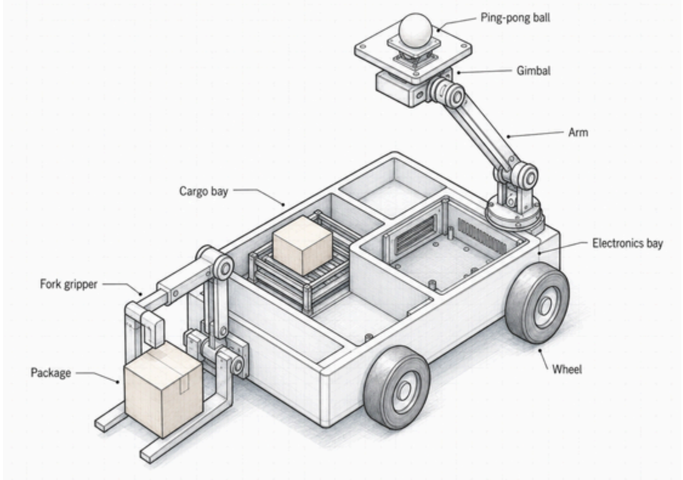
- Line follower
- IMU
- Nucleo F446RE board and PES board
- Servos
- DC Motors with Encoders

CONCEPTS

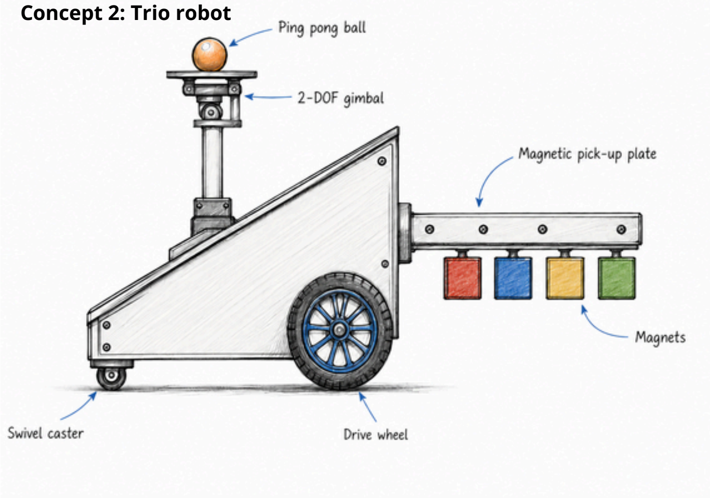
Concept 4: Quad wheel lift



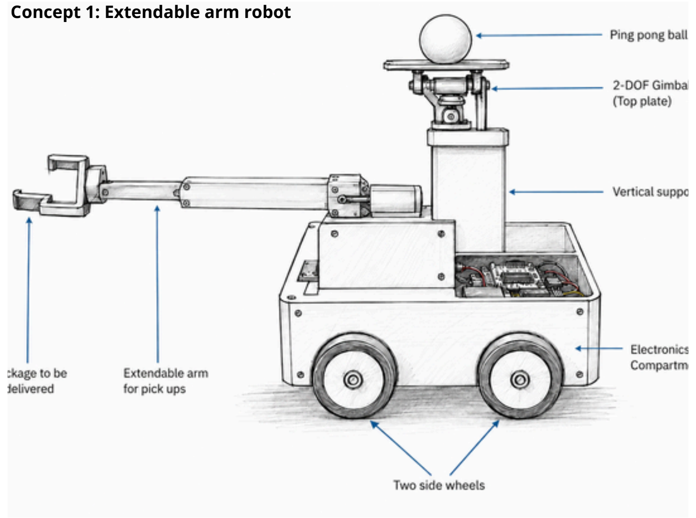
Concept 3: Pickup rover



Concept 2: Trio robot



Concept 1: Extendable arm robot

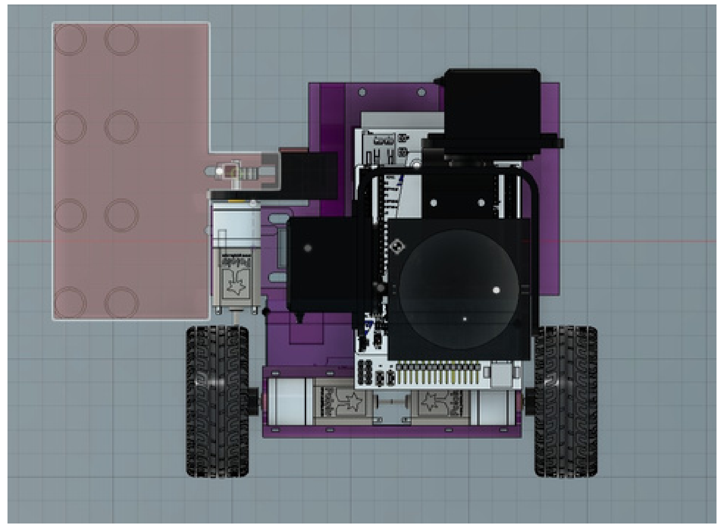


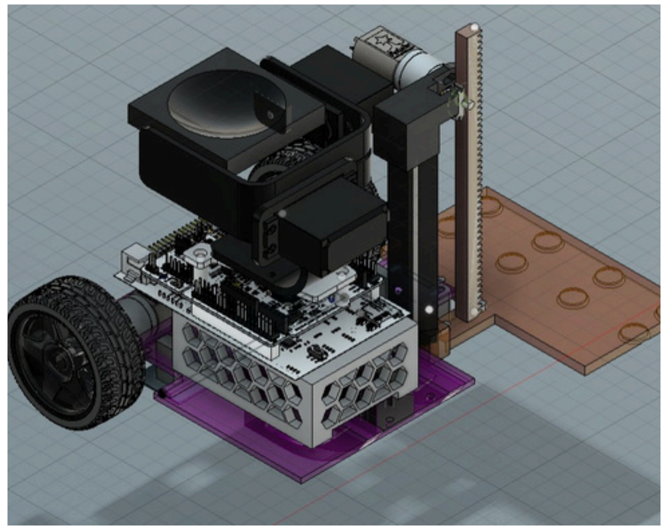
CONCEPT SCORING

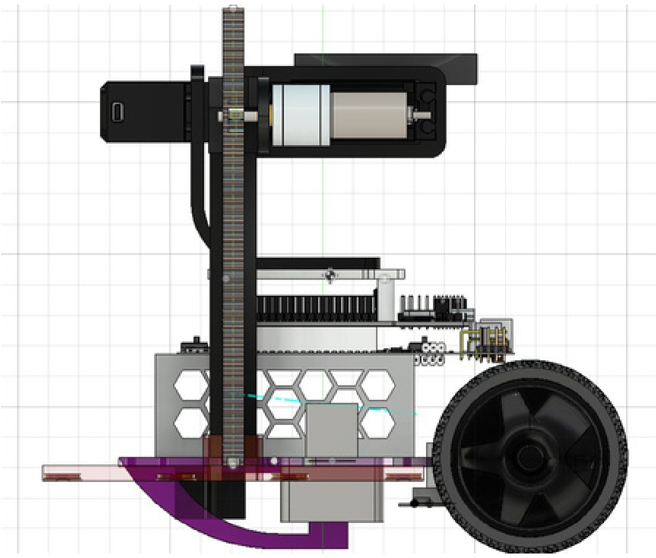
		Concepts									
		Concept1(Extendable arm robot)		Concept 2 (Tria robot)		Concept3 (Pick-up Rover)		Concept 4 (Quad-Wheel lift)		Concept 5 (Point dexter)	
Selection Criteria	Weight (%)	Rating	Weighted score	Rating	Weighted score	Rating	Weighted score	Rating	Weighted score	Rating	Weighted score
Hardware design complexity	13	2.2	0.286	3	0.39	1.2	0.156	3	0.39	1.7	0.156
Software design complexity	13	3.6	0.468	2.3	0.299	1.9	0.247	2	0.26	2.8	0.273
Material availability	10	3.6	0.36	3.7	0.37	2.3	0.23	3.4	0.34	4	0.4
Robustness	10	3	0.3	2.5	0.25	3.3	0.33	3.4	0.34	3.7	0.32
Right package delivery	17	4	0.68	3.2	0.544	2.2	0.374	3.2	0.544	4.7	0.748
Balancing of ping pong ball	17	2.1	0.357	2.2	0.374	2.3	0.391	3.7	0.629	3.5	0.51
Successfully traversing obstacle	17	1.5	0.255	2.1	0.357	3.4	0.578	2.6	0.442	3.2	0.51
Aesthetics	3	2.7	0.081	1	0.03	2.5	0.075	2.5	0.075	2.5	0.075
Total score		2.787		2.614		2.381		3.02		2.992	
Rank		3		4		5		1		2	
Continue?		No		No		No		Combine		Combine	

Rating: 5-1
5: Excellent
4: Good
3: Acceptable
2: Poor
1: Very poor

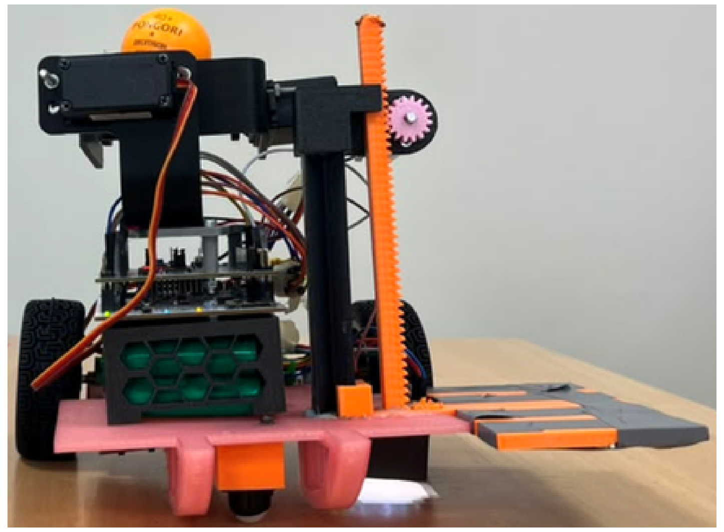
3D MODELS

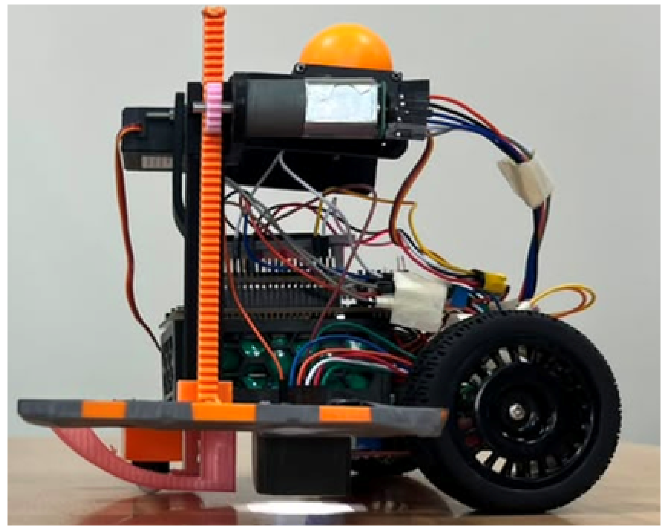


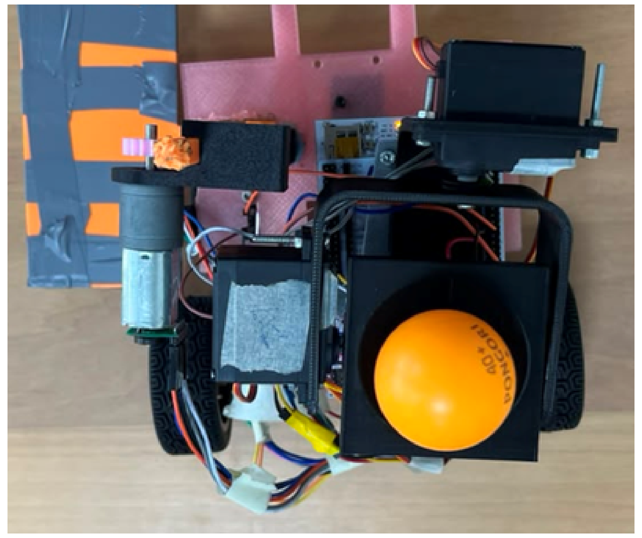


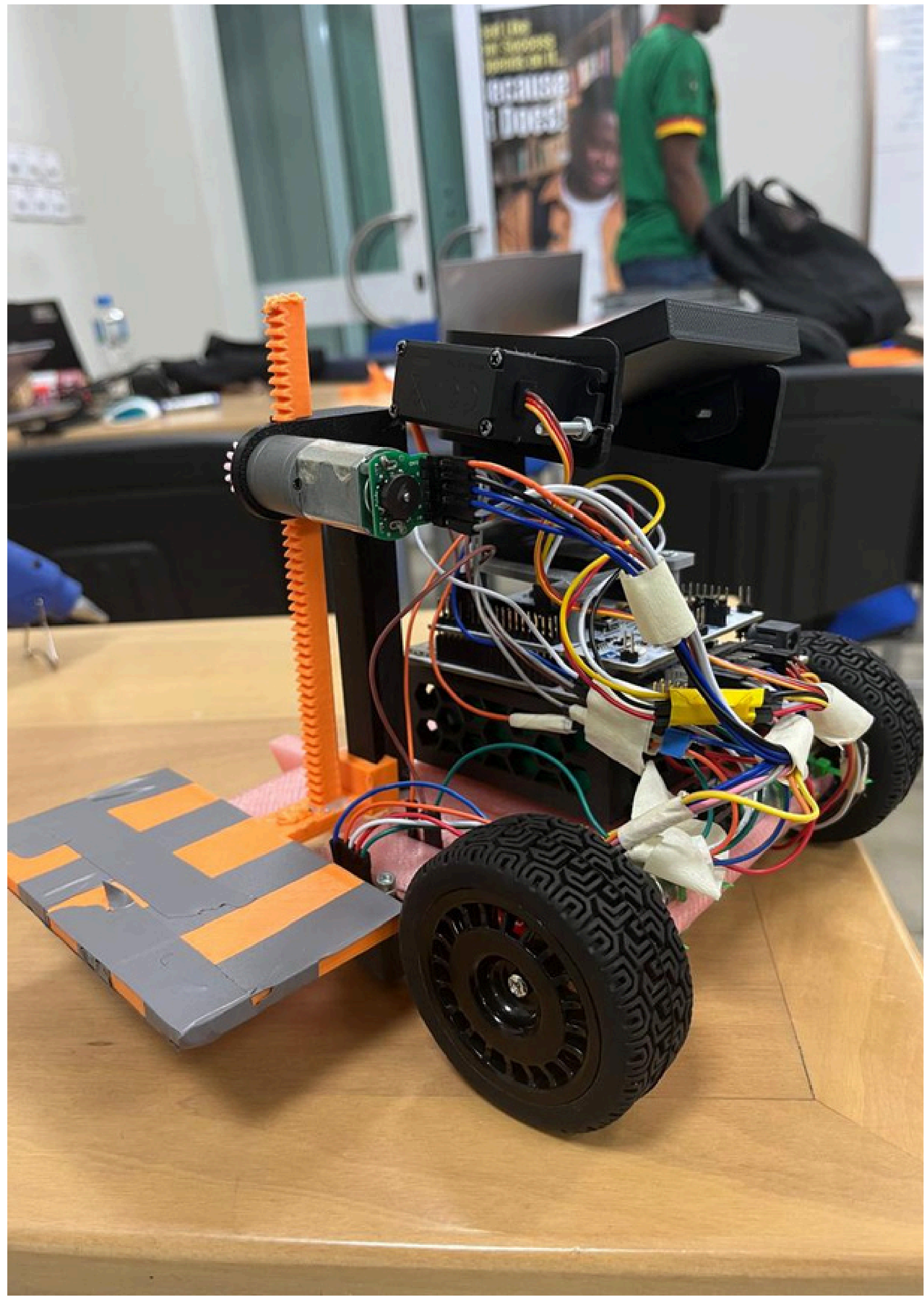


ROBOT VIEWS

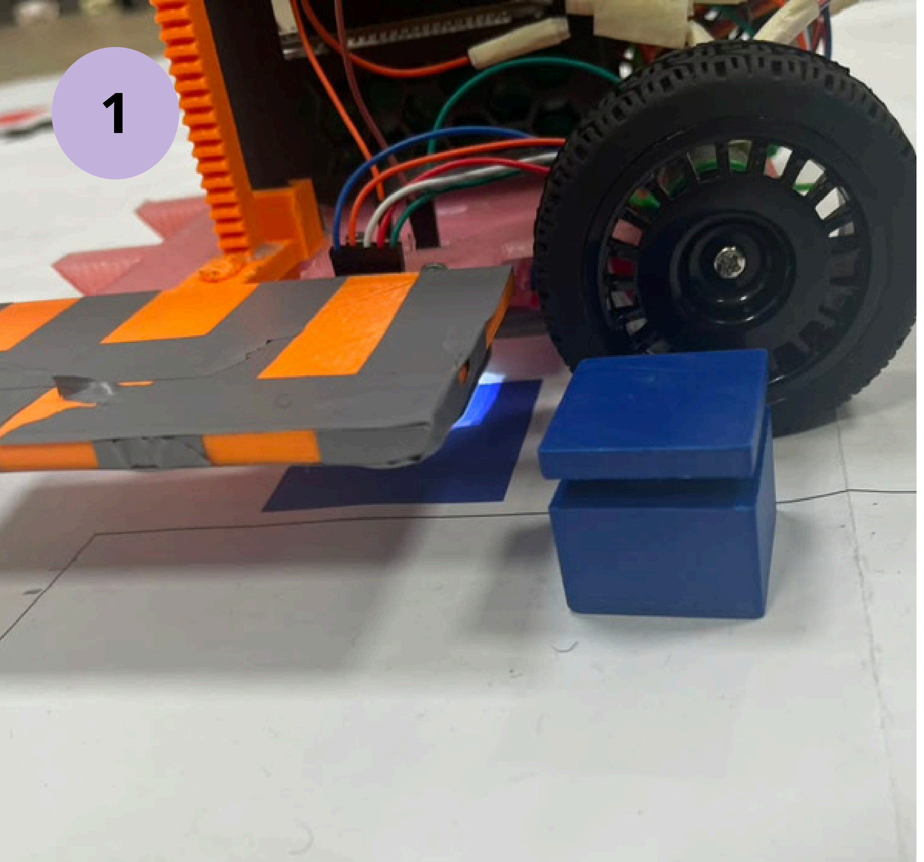






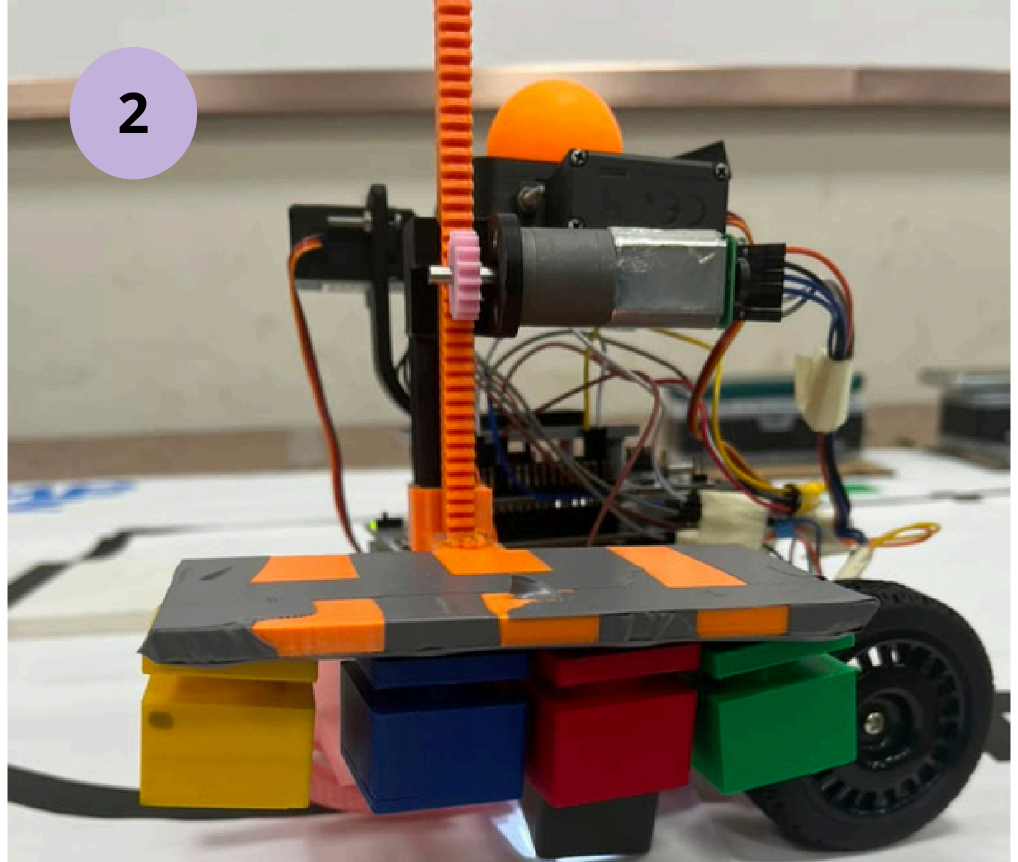


FINAL PROTOTYPE



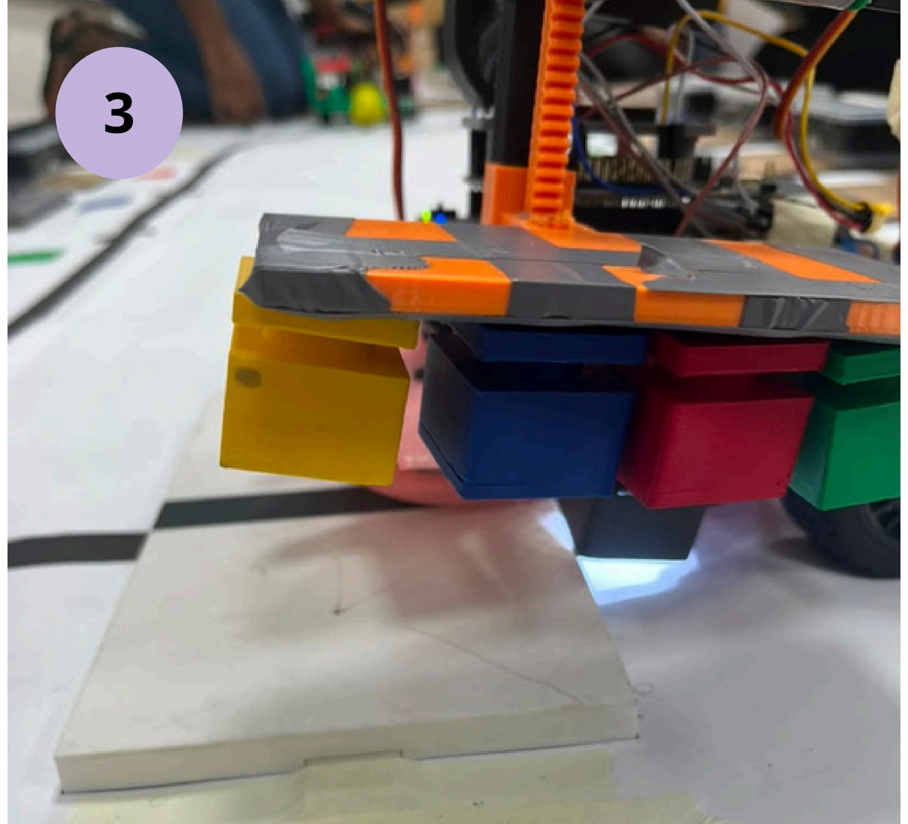
1

Robot sensing blue colour with colour sensor



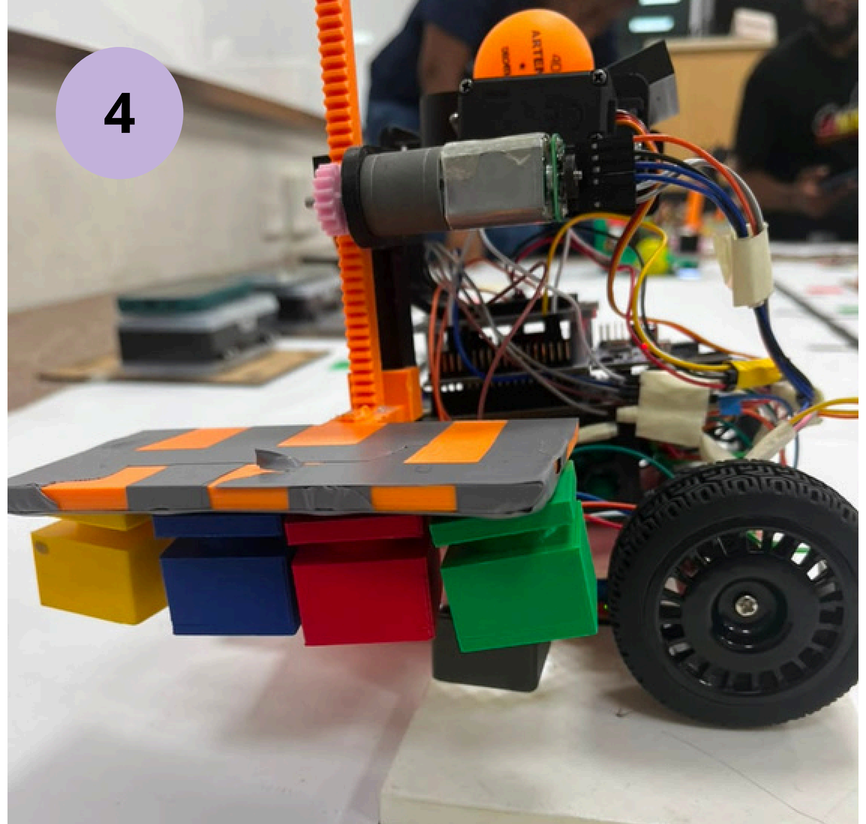
2

Robot with packages about to follow the curve



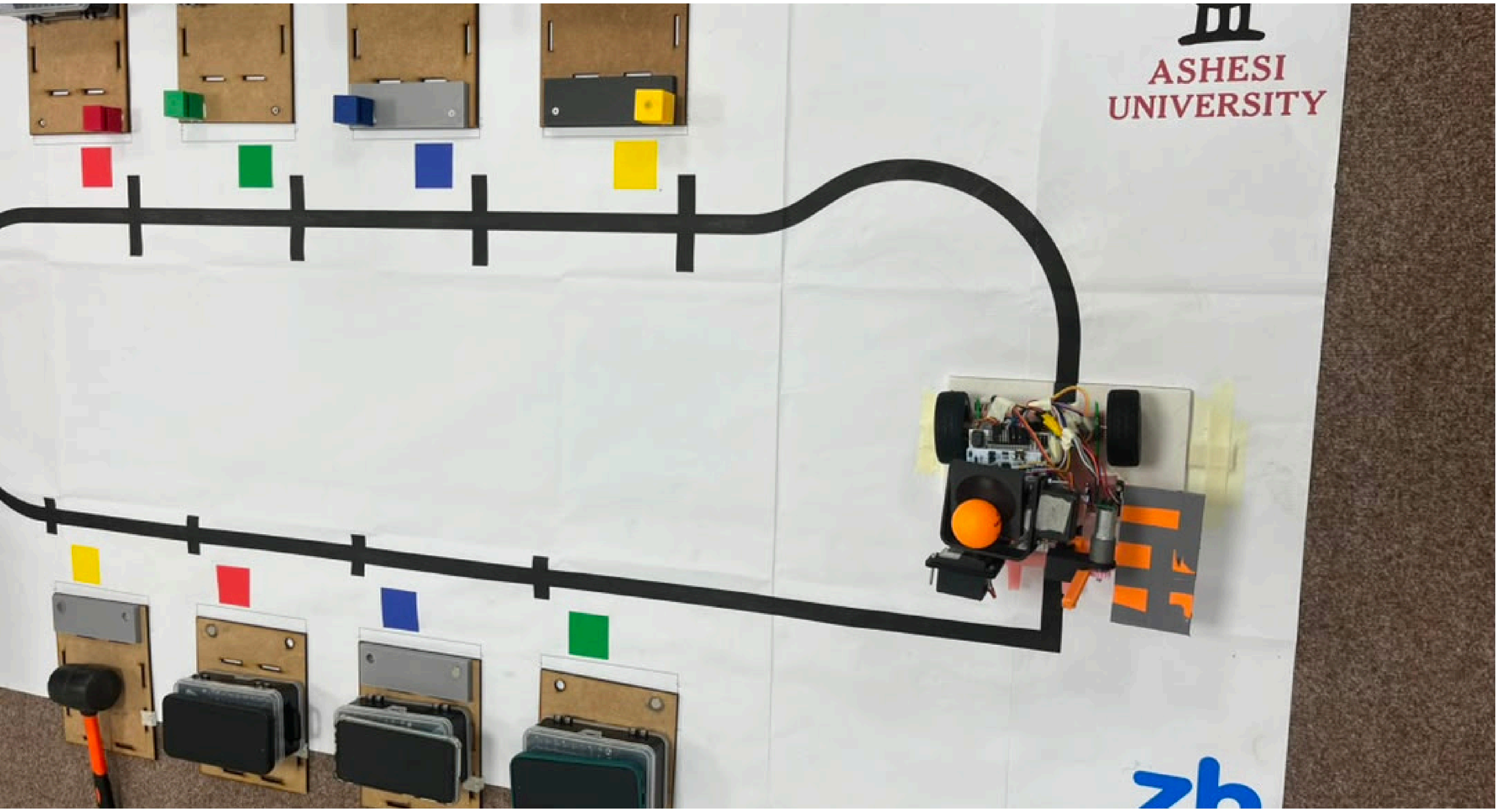
3

Robot with packages about to cross the 10mm step



4

Robot with packages crossing the 10mm step



Robot with packages and ball traversing the obstacle

Unique Selling Points

- Our design is simple
- Our solution is efficient at performing tasks
- Our design is lightweight

Lessons

- Team work is very important
- Systems thinking is essential
- Removal or addition of a part impacts the working of the system.